

Will it sell?

A system that estimates how well a brand-new product would be received — before it ever launches — by learning from real reviews of products like it.

THE BIG IDEA

Judging a product that doesn't exist yet

Before a company builds a new pair of headphones or an ice maker, it wants to know: will people like it? Normally that's a guess. Our system turns the guess into an informed estimate.

It does this the way an experienced buyer would — not by trusting the marketing, but by remembering how thousands of *similar* real products were actually received once people used them. It reads the real reviews of those products, learns what separates the winners from the also-rans, and then judges your new idea against that hard-won experience.

You describe a product with its specs and price. The system gives back a **viability score**, the **strengths and risks** behind that score, and — importantly — an honest **confidence level** that drops when your description doesn't give it enough to go on.

HOW IT WORKS

Five steps, start to finish

The whole system is a pipeline. The first three steps happen once, ahead of time — that's the "learning." The last two happen every time someone asks about a new product.

1

Gather real products and their reviews

We collected **640 real products** across five categories — wireless headphones, bluetooth speakers, smartwatches, power banks, and ice makers — along with about **43,600 genuine customer reviews**.

2

An AI reads every review and scores its "aspects"

A language model reads each review like a panel of careful readers and rates the specific things it mentions — sound quality, value for money, durability, and so on. The strict rule: **if a review doesn't mention something, the AI must say "not mentioned"** rather than invent a score. Honesty about silence is what keeps the data trustworthy.

3

A model learns what success looks like

A machine-learning model (XGBoost) studies all those products and learns the patterns: which combinations of aspect scores tend to go with products that did well, judged **within each category** so a popular speaker is compared to speakers, not to ice makers.

4

For a new product, the AI estimates its aspects from the specs

Your new product has no reviews yet. So the AI acts like a seasoned critic estimating a dish from its recipe: it reads your specs and price and predicts how buyers would likely rate each aspect, anchored to what's normal in that category. We call this the **bridging** step.

5

The model predicts viability — and explains itself

Those estimated aspects go into the trained model, which returns a viability score. It also names the **top strengths and top risks** that drove the score, and a **confidence level** that reflects how much the specs actually told it.

What we mean by "success"

We can't see how many units a product sold — that data isn't public. So we built a stand-in from what we *can* see: a product's **success score** blends how many reviews it has (40%), its average rating (35%), and how fast reviews arrive (25%).

This measures **engagement and popularity**, not sales. A product everyone talks about and rates highly scores well. It's a reasonable proxy, but it's a proxy — and we're upfront about that everywhere it matters.



Why this matters

Because the target is popularity rather than real sales, the system predicts "how well-received would this be," not "how many will sell." Everything below should be read with that in mind. Swapping in a real sales signal is the single biggest future improvement.

THE VOCABULARY

The nine things it judges

Every product is broken down into the same nine aspects. Six are universal; three are specific to physical goods.

Value for money

Utility does its core job

Ease of use

Reliability

Design appeal

After-sales support

Build quality

Durability

Repairability

HOW GOOD IS IT

Reliable at ranking, honest about the rest

The most important test: take products the system has *never seen*, predict them from specs alone, and check the predictions against what really happened. We ran the strictest version — training only on older products and predicting *newer* ones, exactly like forecasting the future.

0.90

Spots the hit

Given a future hit and a future flop, it scores the hit higher about 90% of the time. (This is the "AUC.")

0.55

Ranks the future

Its ordering of unseen, later products matches reality well. (0 = random, 1 = perfect. This is "Spearman correlation.")

9 + 2

Always explains

Every score comes with the top strengths and risks behind it — never a black-box number.

We measured it three ways. All agree the system is trustworthy for **ranking** products (which will do better), not for exact percentages.

TEST	RANKING (P)	SPOTS THE HIT (AUC)	WHAT IT SHOWS
Cross-validation	—	—	Fits the known data well (R^2 0.60 on aspects alone)
Random holdout	0.44	0.83	Works on unseen products
Forward-in-time	0.55	0.90	Works predicting the future

Out-of-sample results. Higher is better; both scales run 0 (useless) to 1 (perfect).



Where it's weak — and we say so

The system is strongest for smartwatches, speakers, and headphones, and weakest for ice makers — its newest and thinnest category, where forward accuracy drops close to zero. It's good at ordering products, not at pinning down an exact score, so we present results as a range and a rank, never as false precision.

WHAT MAKES IT CREDIBLE

The details that earn trust

A prediction is only as good as its honesty about its own limits. Several parts of the system exist purely to keep it honest.

It knows what it doesn't know

Confidence is weighted by how reliably each aspect can actually be judged from specs. Vague descriptions get low confidence — and below a threshold, the system refuses to show a score at all rather than bluff.

It understands brands

A new product from an established brand and one from an unknown startup are judged differently — the model learned that brand track record affects trust, service, and reliability, instead of treating every maker the same.

It separates skill from luck

Average rating is part of our success measure, so a model using it looks artificially good. We report a stricter "aspects-only" model that ignores such shortcuts — the honest measure of what the review-reading actually adds.

It shows its reasoning

Every prediction lists the specific aspects pushing the score up or down (via a method called SHAP), so a designer sees exactly what to improve — not just a verdict.

THE PAPER'S CLAIM

What this contributes

Most published work in this area classifies the sentiment of reviews that *already exist*. Our contribution is different: a two-stage system that combines **AI review-reading** with a **gradient-boosted model** to estimate viability for a product that has *no* reviews yet — and does so **explainably and with calibrated, honest confidence**.

The defensible headline is not "we predict sales." It is: *aspect signals extracted by a language model carry real, measurable predictive signal about market reception, and that signal generalizes to unseen and future products* — demonstrated with proper out-of-sample and forward-in-time validation, and reported alongside its limits.

WHAT'S NEXT

This is the defensible first version

This copy is built to be honest and well-evidenced — the version the capstone and paper stand on. The next phase is about raising accuracy, and we already know the two biggest levers:

A real sales signal

Replacing the popularity proxy with something like sales rank would make the target — and the claim — meaningfully stronger.

Products that failed

Today's data only includes products that already gained traction. Adding low-traction and just-launched products would teach the model what failure looks like, which matters most for brand-new brands.

IN PLAIN WORDS

Glossary

Aspect

the vocabulary

One specific quality of a product — value, durability, ease of use. The system judges nine of them.

Bridging

step 4

Estimating a not-yet-launched product's aspect scores from its specs, since it has no reviews to read.

Success score

the target

Our stand-in for market success, built from review volume, average rating, and review speed. A popularity proxy, not sales.

XGBoost

the model

A well-proven machine-learning method that learns patterns from tables of numbers. It does the actual predicting.

SHAP

the explanation

A technique that breaks a prediction into "this aspect pushed the score up, that one pulled it down."

Confidence

the honesty dial

How much to trust a given prediction. It falls when the specs are thin, and the system stays silent when it's too low.

R^2

a fit measure

How much of the variation the model explains on known data. 0 is nothing, 1 is perfect.

Spearman ρ

a ranking measure

How well the model's ordering matches reality. 0 is random, 1 is a perfect ranking.

AUC

a sorting measure

Pick a hit and a flop at random — how often does the model score the hit higher?

0.5 is a coin flip, 1 is perfect.

Market Success Viability Predictor · a capstone in explainable machine learning

5 categories · 640 products · ~43.6k reviews · AI aspect extraction → XGBoost → SHAP